BOMSC-Net: Boundary Optimization and Multi-Scale Context Awareness Based Building Extraction From High-Resolution Remote Sensing Imagery

Yuan Zhou, Zhanlong Chen, Bin Wang, Shuangjiang Li, Hao Liu, Daozhu Xu, and Chao Ma

***Abstract*— Automatic building extraction from high-resolution remote sensing imagery has various applications, such as urban planning and land use management. However, the existing methods for automatic building extraction focus on regional accuracy but not on boundary quality. As a result, these building extraction methods yield discontinuous results due to tree and shadow occlusions and complex building roof materials. In this article, BOMSC-Net, which can consider boundary optimization and multi-scale context awareness, is proposed to address these two problems. BOMSC-Net is an encoder-to-decoder network composed of a basic network, a multi-scale context-aware module (MSCAM), and a direction feature optimization module (DFOM). The basic network learns to predict building segmentation maps through attention gates (AGs) that connect low-level feature jumps to advanced semantics. MSCAM is used as a bridge, which uses graph reasoning blocks (GRBs) to solve intra-class discontinuities. Finally, the DFOM constructs a direction field (DF) and then uses direction information for boundary correction to achieve fine building segmentations. Experiments were conducted on the open Massachusetts building dataset, WHU aerial building dataset, and Inria aerial image labeling dataset. The quantitative and qualitative analyses demonstrate that the proposed BOMSC-Net can help building extraction overcome interruptions caused by shadows and occlusions, extract buildings with different scales and materials, and realize boundary optimization and complete segmentation of buildings.**

***Index Terms*— Boundary optimization, building extraction, direction field (DF), graph reasoning block (GRB), multi-scale context-aware module (MSCAM).**

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I. Introduction

As CENTRAL locations of human activities, cities are of great significance, so realizing effective analysis and evaluation of a city's development process is significant for guiding sustainable development of the region [1]. As one of the main constituent elements of a city, buildings are inevitably a focus of researchers. Remote sensing imagery is currently the mainstream source for obtaining ground feature information. Therefore, effective extraction of buildings from remote sensing imagery is vital to applications such as urban planning, land use management, resource allocation, and convenience services [2]. Although high-resolution remote sensing imagery is rich in feature information, complex spectrum, structures, and texture information of features also present severe challenges to accurately extracting buildings.

In the past, the use of image-processing-based methods enabled the possibility of building extraction. These research methods can be divided into methods based on building features [3], [4] and methods based on auxiliary information [5]. The methods based on building features use spectrum, texture, geometry, and other building characteristics to extract the building. Among them, corner detection operators [6], edge detection operators [7], image transformation [8], and histograms [9] are widely used in building feature extraction. However, because trees and shadows often block buildings, various detection operators cannot fully use the general geometric information. Some researchers construct a template library based on the prior shape of the building and introduce it into the active contour model to constrain the evolution of the segmentation curve [10], [11]. However, this method is limited regarding complex and diverse building shapes in a large area. The methods based on auxiliary information combine multi-source remote sensing data [12], [13], GIS data [14], and other rich building feature information to achieve accurate extraction. However, these methods are often time-consuming and extremely costly. This method of relying on expert experience and manually designed features is only suitable for some specific tasks. It is currently difficult to meet research needs through this approach [15], [16].

In recent years, owing to the advantages of being able to learn discriminatory features actively, deep learning has been

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widely used in the field of remote sensing, including change detection [17], [18], scene recognition [19], [20], and land use classification [21], [22]. Among them, the most typical method is the convolutional neural network (CNN) [23]. CNN can obtain hierarchical data representation. High-level feature representation relies on low-level feature representation. With the construction of a multi-layer convolution operation from shallow to deep, the features with more advanced semantic information are extracted gradually. Then, the fully convolutional network (FCN) [24] was proposed, which successfully broke the patch-wise barrier by removing the fully connected layer and achieved pixel-wise performance. This is essential for this data-intensive task of building extraction.

Inspired by the above, the method based on deep learning offers excellent performance in building extraction [25], [26]. Furthermore, some studies improved the FCN framework to obtain high-precision building segmentation results [27]–[29]. However, accurate extraction of buildings from remote sensing images is often affected by many objective conditions. On one hand, due to the spatial diversity of the building itself, its shape is complex, the size differs, and the roof material may be different; on the other hand, the imaging conditions of remote sensing images are disturbed by the environment, and buildings are often blocked by shadows and trees. These effects make it difficult for the divided buildings to maintain their unique morphological characteristics and integrity.

The boundary can describe the morphological shape of the building well [30]. Therefore, some studies achieve fine building segmentation by enhancing the description of boundary information or using postprocessing techniques. For example, Jung *et al.* [31] proposed a method composed of holistically nested edge detection and a boundary enhancement module, which extracts edge features and assists in generating building segmentation results with enhanced boundary information. Ma *et al.* [32] constructed an encoder with local and global feature extraction to fully describe buildings of different shapes. Although this method can obtain useful information that pays more attention to the boundary, adding a complex network structure and additional auxiliary data also increases consumption. Postprocessing technology is a commonly used method to solve blurred boundaries of building segmentation. Some filters, such as a guided filter [33], morphological filter [34], and other methods, optimize building boundaries and improve boundary regularity. In addition, some methods try to refine the results of FCNs using probabilistic graph models, such as fully connected conditional random fields (CRFs) [35]. For example, Zhu *et al.* [36] proposed to combine segmentation-prior CRFs to carry out building extraction work based on a multi-scale semantic segmentation network (D-LinkNet). Sun *et al.* [37] proposed a new time-efficient end-to-end CRF model to reduce time consumption. Although the postprocessing method has been greatly improved, its computational cost is high, and the time consumption problem cannot be completely solved. To further improve building shape, a multi-task framework is applied to building extraction. Wu *et al.* [38] proposed a boundary regulated network, which uses an improved U-Net to perform multi-task learning to obtain segmentation maps and edge contour maps, to obtain the results of boundary optimization. However, there is no direct connection between tasks, and the burden of model training will continually increase.

The result of building extraction is also reflected in completeness. Some studies have attempted to capture the contextual information of multi-scale objects by constructing a wider [39], [40] and deeper [41] network. Most of these methods use pyramid structures or convolution operations with different receptive fields to fuse features of different scales and resolutions to enhance contextual information learning of buildings. Sun *et al.* [42] proposed using an improved encoder with multi-scale convolution to extract depth features of buildings. Some researchers used spatial pyramid pooling operations to connect the encoder and decoder features hierarchically to extract multi-scale building information [43], [44]. Although feature fusion of different resolutions in the process of feature extraction can enhance the contextual information of the building, the location information of the building may be lost, leading to blurry boundaries or excessively smooth corners. Introducing skip connections [45] in the decoder and capturing long-distance dependencies in space or channel through the attention mechanism [46], [47] represent important ways to solve the lack of spatial information. Yang *et al.* [48] combined a dense network and attention mechanism to form a new dense attention network. The purpose is to enhance low-level feature information and prevent excessively redundant low-level features from being introduced into the model. Pan *et al.* [49] combined space and channel attention mechanisms to construct a new generative adversarial network to achieve fine segmentation of buildings. The incomplete segmentation caused by the difficulty of capturing long-distance area has not yet been solved.

However, unclear boundaries or variable intensity lead to the lack of feature correlation between pixels. The global correlation between long-distance regions cannot be guaranteed. There are still problems of blurred boundaries and incomplete extraction in building extraction. To solve the above problems, BOMSC-Net is proposed to achieve fine segmentation of multi-scale buildings. On one hand, with the introduction of attention gates (AGs) in the basic network, low-level features are connected to advanced semantic information to realize attention to significant areas and suppress irrelevant background. In addition, a direction feature optimization module (DFOM) is used to further perceive the direction information of buildings to refine the results of building boundary segmentation. On the other hand, a multi-scale context-aware module (MSCAM) is introduced. Instead of stacking deep convolution operations to implicitly construct the global context, the module complements the global information of buildings through graph reasoning blocks (GRBs). Experiments were conducted on the open Massachusetts building dataset, WHU aerial building dataset, and Inria aerial image labeling dataset. Quantitative and qualitative analyses demonstrate that BOMSC-Net achieves outstanding performance in terms of generalization and robustness.

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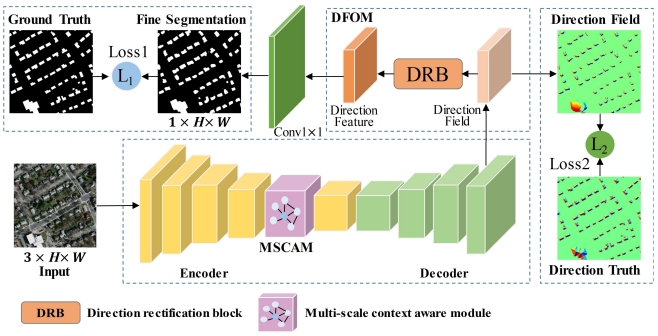

Fig. 1. Schematic of the proposed segmentation framework. Loss1 adopts the general binary entropy loss L_{BCE} and L_{dice}. Loss2 uses direction field loss L_{DF}.

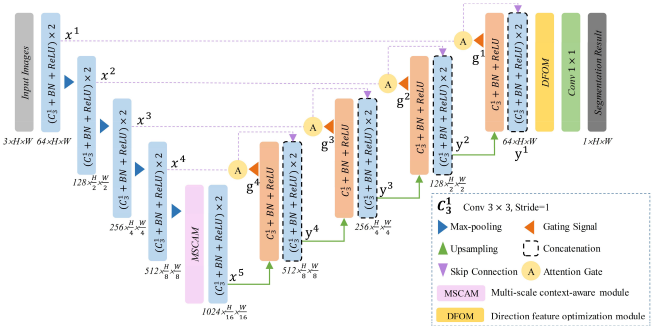

Fig. 2. Detailed process of the basic network. $x^l (l \in [1, 2, 3, 4, 5])$ represents the feature map obtained by the encoder, and its size is $2^r \cdot 64 \times (H/2^r) \times (W/2^r) (r \in [0, 1, 2, 3, 4])$. y^l represents the feature map by the decoder. l represents the number of layers, and r is the ratio. Note that x^5 and y^5 are the same feature map, and only x^5 is marked in the figure. g^l represents the gating signal.

The main contributions of this study are as follows.

1) This study proposes a novel segmentation network framework, BOMSC-Net, to achieve fine building segmentation in high-resolution remote sensing.
2) The MSCAM, which uses pooling operations of different kernel sizes in the branch path and then a parallel graph reasoning module to realize context inference of multi-scale buildings, is designed.
3) The DFOM aims to use the directional relationship between pixels to guide fine segmentation of building boundaries.

The remainder of this article is organized as follows. Section II introduces the details of the proposed BOMSC-Net. Section III describes the implementation details of the experiments. Section IV displays the experimental results and discussion content. Section V provides the conclusion.

II. Methodology

This study proposed a multi-module CNN network framework, BOMSC-Net, to extract buildings from remote sensing images. The network comprises the basic network, the MSCAM, and the DFOM. As shown in Fig. 1, the basic network uses multi-layer convolution operations on the input RGB images to process local information layer by layer to obtain higher dimensional semantic information.

In addition, the basic network connects low-level features to high-level semantic information through AGs to direct attention to significant areas of buildings and to suppress irrelevant backgrounds. In particular, an MSCAM is embedded between encoding and decoding to realize the inference of global context information to better express multi-scale features of buildings. To further refine the final segmentation results, this study designs a DFOM to dig deeper into the spatial relationships between each pixel and uses the directional spatial relationship as a guide to achieve refined segmentation of the network results.

A. Basic Network

The basic network of this study is a universal encoding–decoding end-to-end structure. The basic network takes RGB image $I \in \mathbb{R}^{3\times H\times W}$ as input, where $H \times W$ refers to the size of the input image, as shown in Fig. 2. The encoder is composed of multiple consecutive convolution and down-sampling operations. The pixels are separated to obtain their high-level semantic information at the cost of increasing the channel dimension. First, I can obtain low-level features x^1 through convolution operations (the kernel size is 3, and the stride is 1), linear transformation (batch normalization), and nonlinear activation function [rectified linear unit (ReLU)] in turn. Applying a max-pooling operation to x^1 can achieve down-sampling to obtain features with more local information. This feature will be used as the input feature of the next layer, and a similar operation of the previous layer will be repeated. There are five layers in encoding. Using l to represent the number of layers, each layer can obtain feature maps x^l $(l \in [1, 2, 3, 4, 5])$ with different sizes and channel sizes.

The decoder is realized by an up-sampling operation combined with multiple consecutive convolution operations. The decoding process consists of four layers. The feature map y^l obtained at each layer will undergo up-sampling and convolution operations to gradually restore image features. Specifically, this study uses the nearest neighbor interpolation algorithm for up-sampling, and the scale factor is set to 2.

In particular, we use a special skip connection to redistribute the weights of different regions in low-level feature of the encoding stage. Weight redistribution is realized through AGs [46], which prompt the network to pay attention to the area of interest, suppress the irrelevant background, and learn more local information about the building. Fig. 3 shows the implementation process of AGs. The output of AGs is element-wise multiplication of input feature maps and attention coefficients.

First, up-sampling and convolution operations are performed on y^{l+1} to obtain the gating signal g^l. The formula is expressed as

$$g^l = \text{Conv}\left(\Phi(y^l)\right) \tag{1}$$

where Conv denotes the convolution operation (the kernel size is 3, and the stride is 1), and Φ denotes the linear transformation (batch normalization) and nonlinear activation function (ReLU).

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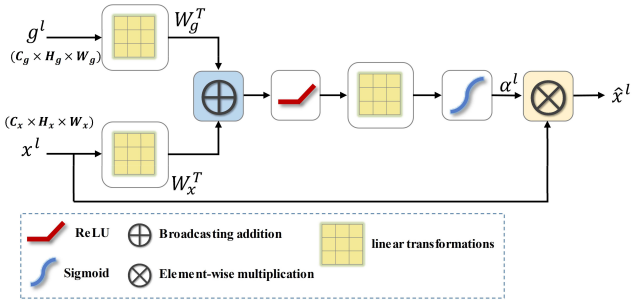

Fig. 3. Detailed process of AG. The feature map x^l from the l layer encoding undergoes a series of operations to obtain the attention coefficient $\alpha^l \in [0, 1]$. α^l can identify salient image regions and suppress irrelevant backgrounds. Then, the feature map $\hat{x}^l$ after attention enhancement can be obtained.

Second, the gating signal g^l and the feature map x^l from the corresponding layer of the encoding are used to calculate the attention coefficient α^l. The following formulas can express it:

$$\alpha^{l-1} = \sigma\left(\psi^T\left(R\left(W_x^T x^{l-1} + W_g^T g^l + b_g\right)\right) + b_\psi\right) \quad (2)$$

where W_x^T, W_g^T, and ψ^T denote the linear transformations, and b_g and b_ψ denote the bias terms. Both W_x^T and W_g^T are achieved through a convolution operation (the kernel size is 1, and the stride is 1) and a batch normalization. R denotes the ReLU, and σ corresponds to the Sigmoid activation function.

Then, by performing element-wise multiplication on α^l and x^l, a new feature map $\hat{x}^l$ with attention weight can be obtained. The specific formula is as follows:

$$\hat{x}^l = x^l \cdot \alpha^l, \quad \alpha^l \in [0, 1]. \quad (3)$$

Finally, the feature map $\hat{x}^l$ after attention enhancement and the decoding feature map are aggregated by concatenation operation. The decoding feature map y^l is obtained at the output of layer l by sequentially applying multiple convolution operations (the kernel size is 3, and the stride is 1) and a linear transformation (batch normalization) followed by a nonlinear activation function (ReLU). y^l will be used as the input feature map of the decoder of the next layer, and the above process will be repeated.

Note that this study embeds an MSCAM between the encoder and the decoder of the network and embeds a DFOM in the penultimate layer of the decoding part. For details, see the next section. Finally, the last layer of the network uses a 1×1 convolution operation (the kernel size is 1, and the step size is 1) to replace the role of the fully connected layer. Then, the final building extraction result is obtained through a Sigmoid activation function.

B. Multi-Scale Context-Aware Module

To enhance the network's ability to reason regarding contextual information, thereby capturing feature maps with global contextual information, this study designs an MSCAM to aggregate semantic information between remote regions of different scales. In Fig. 4, paths with different pooling sizes are constructed hierarchically into a feature map. GRBs are applied to each layer of paths to achieve efficient global information inference.

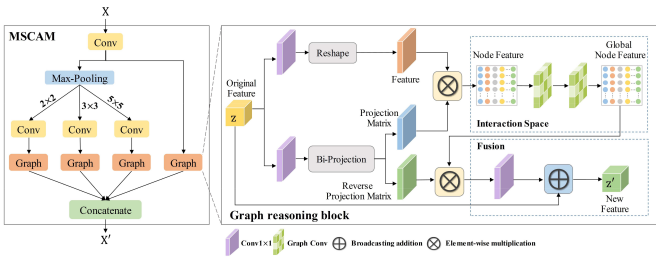

Fig. 4. Illustration of MSCAM. In particular, the figure shows the structure of the GRB in detail.

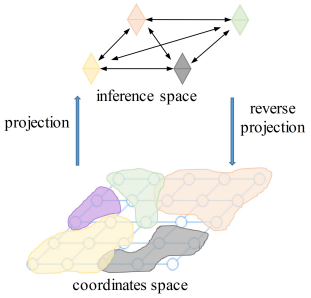

Fig. 5. Schematic of the inference space.

The MSCAM is divided into four paths. First, we perform a 3×3 convolution operation on the input feature map X. Then, three paths must undergo max-pooling operations with different kernel sizes. Specifically, the kernel sizes for max-pooling operations are set to 2×2, 3×3, and 5×5. To preserve the original features, the remaining path is not pooled. We use different receptive fields to capture and learn multi-scale building information. Second, these features are fed into the GRBs to better learn the object's remote semantic context. Finally, the feature maps of each branch path after passing through the GRB are unified through bilinear interpolation, and then the final output feature map X' is obtained through concatenation.

Graph reasoning has been proven to be an effective method of relationship reasoning for obtaining object relationships between remote regions [50], [51]. To obtain global context information for buildings, this study projects a feature map into an inference space that can interactively describe regional relationships to capture the distance between disjoint regions with arbitrarily shaped relationships.

Different from the regular coordinate space $\mho = \{1, \ldots, H \times \{1, \ldots, W$, the latent inference space H expresses the relationship between regions through nodes, edges, and adjacency matrices. Concretely, the convolution is performed over a regular nearest neighbor graph defined by an adjacent matrix $A \in \mathbb{R}^{L \times L}$, where $A_{ij} = 1$ if regions i and j are spatially adjacent; otherwise, $A_{ij} = 0$. The edges of the graph encode spatial proximity, and its node stores the feature for that location, as shown in Fig. 5.

The detailed description of the GRB is as follows.

Step 1: For the original feature map $Z \in \mathbb{R}^{C \times H \times W}$, C is the channel size of the feature map, and the original regular coordinate space is $\mho = \{1, \ldots, H \times \{1, \ldots, W$. It is necessary to find the projection function $f(\cdot)$ that maps the original feature map to the inference space . In the inference space,

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we can construct a new fully connected graph A_g, where each node stores the new feature as its state. A new feature map $V = f(Z) \in \mathbb{R}^{C\times N}$ is obtained by transforming the projection function, where N is the number of nodes of feature Z in the inference space . Specifically, V can be expressed by the following formula:

$$v_i = Zb_i = \sum_{\forall j} z_j b_{ji} \tag{4}$$

where $B = [b_1, \ldots, b_N] \in \mathbb{R}^{H\times W\times N}$ denotes the learnable projection weights, and $z_j \in \mathbb{R}^{C\times 1}, v_i \in \mathbb{R}^{C\times 1}$. In practice, to reduce the input dimension and to increase the capacity of the projection function, we implement the function $f(Z)$ as $f(\emptyset(Z; W_\emptyset))$ and $B = \theta(Z; W_\theta)$. We model $\emptyset(\cdot)$ and $\theta(\cdot)$ by two convolution layers. $W_\emptyset$ and W_θ are the learnable convolution kernels of each layer.

Step 2: To capture the interaction between features of the corresponding nodes, the contextual relationship between each pair of nodes is modeled and inferred by applying a general graph convolution. The formula can be described as follows:

$$V' = GVW_g = \big(\big(1 - A_g\big)V\big)W_g \tag{5}$$

where G denotes a graph of the original, and W_g denotes the weight parameter matrix. V' is a new feature obtained through graph convolution inference.

Step 3: It is also necessary to perform back projection to reverse the features in the inference space back to the original coordinate space. Then, the projected features can be effectively integrated with the original framework network to provide better decision-making for subsequent network learning. The specific formula is as follows:

$$Z' = G\big(V'\big), \quad V' \in \mathbb{R}^{C\times N}, \ Z' \in \mathbb{R}^{C\times H\times W} \tag{6}$$

where Z' refers to the features of the coordinate space that are reverse-mapped through the projection function $g(\cdot)$. Similar to (4), linear projection is used to formulate $g(V')$

$$\mathrm{z}_i{}' = d_i V' = \sum_{\forall j} d_{ij} v_j{}'. \tag{7}$$

The feature v'_j of node j is assigned to z'_i weighted by a scalar d_{ij}.

C. Direction Feature Optimization Module

The current building segmentation methods based on deep learning lack global control over the correlation between pixels, resulting in the inability of segmented buildings to maintain their morphological shapes. To further refine the boundary information of the building, this study uses DFOM to explore the spatial relationships between pixels and strengthen the semantic associations between pixels to achieve high-precision segmentation. As shown in Algorithm 1, this study depicts the DFOM algorithm in the form of pseudocode.

1) Direction Field: The direction field (DF) of each pixel p is defined as a two-dimensional unit vector pointing from its nearest boundary pixel b to pixel p [52], [53]. DF provides cues about contour positions and the relative position of each pixel p to its region boundary. In the image domain Ω, for each foreground pixel p located inside the buildings, we find its nearest boundary pixel b located at the boundary of the building. Background pixels represent pixels in non-building areas. Then, we describe the distance and direction information between pixel b and pixel p by constructing the directional vector $\underset{bp}{\longrightarrow}$ from boundary pixel b to pixel p inside the building. At the same time, other vectors that pass through the background pixels are set to (0, 0). Therefore, the boundary and directional characteristics of buildings can be represented by the DF as follows:

$$\mathrm{DF}(p) = \begin{cases} \dfrac{\overrightarrow{bp}}{\overrightarrow{|bp|}}, & p \in \text{buildings} \\ (0,0), & \text{otherwise.} \end{cases} \tag{8}$$

The DF of the network can be learned through a 1×1 convolution operation. Specifically, this study specifies the 64-channel feature map extracted from the decoding layer as the initial segmentation map $F^0(p) \in \mathbb{R}^{C\times H\times W}$. Applying a 1×1 convolution operation to $F^0(p)$ can yield the two-channel prediction direction field $\mathrm{DF}(p) \in \mathbb{R}^{2\times H\times W}$. Note that the direction truth from the ground truth can be efficiently achieved by the distance transform algorithm [54]. For the binary image ground truth, the foreground pixel is 1, and the background pixel is 0. The distance transformation algorithm can quantify the minimum distance between pixels 1 and 0. The result of the distance transformation is a grayscale image, that is, a distance image. The gray value of each pixel in the image is the distance between the foreground pixel and the nearest background pixel. The Euclidean distance used in the calculation by the distance transformation algorithm is specified as

$$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}. \tag{9}$$

2) Direction Rectification Block: The direction rectification block can gradually correct the errors of the initial segmentation map with the help of acquired DF. The direction

Algorithm 1 Direction Feature Optimization Module

Input: Image and ground truth (**I**, **G**)
while in training stage: **do**
  Get the batch of initial segmentation **F** and network's initial parameter θ
  // step1: Obtain the predicted direction field F_{df} and the direction truth R
  $F_{df} \leftarrow$ Conv_1 × 1 (**F**)
  R ← Distance transform (**G**)
  // step2: Direction rectification
  $F^0(x, y), \mathrm{F}^0_{df}(x, y) \leftarrow$ Grid (**F**, F_{df})
  for i = 1, . . . , N **do**
    $F^i \leftarrow \mathrm{Rectify}(F^{i-1}(x, y), F^{i-1}_{df}(x, y))$
    $F^i \leftarrow$ grid_upsample (F, F^i)
  end for
  // Update parameters
  $\hat{\theta} \leftarrow \theta - (\alpha\nabla L_{bce}(\theta, G) + \beta\nabla L_{dice}(\theta, G))$
  $\hat{\theta}_{df} \leftarrow \theta_{df} - (\gamma\nabla L_{df}(\theta_{df}, F_{df}, R))$
end while

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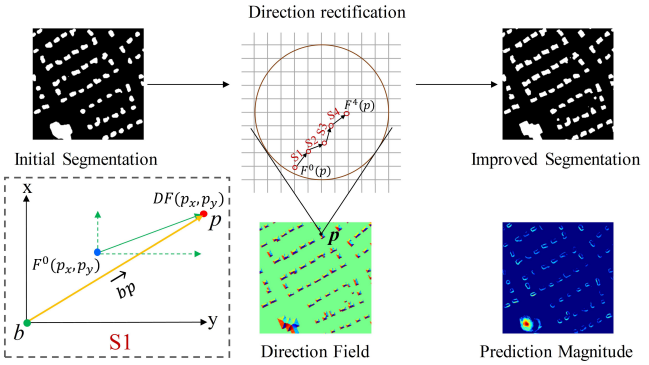

Fig. 6. Illustration of the direction rectification block. The initial segmentation map $F^0(p)$ is gridded. The boundary rectification is performed on $F^0(p)$ using the pixels at the corresponding position in the direction field (DF), and $F^1(p)$ can be obtained. After four corrections, an improved segmentation map $F^4(p)$ is obtained. The lower left corner is a schematic of the correction process of S1. $\underset{bp}{\longrightarrow}$ is a directional vector. $F^0(p_x, p_y)$ is the pixel p predicted by $F^0(p)$. $DF(p_x, p_y)$ is the position of the corresponding pixel p in DF. $F^0(p_x, p_y)$ is corrected once through $\underset{bp}{\longrightarrow}$ so that the building boundary is optimized once. S2, S3, and S4 are the same as S1.

rectification block iteratively corrects the pixel close to the boundary pixel through features in the center area.

First, grids $F^0(p)$ and $\mathrm{DF}(p)$ facilitate the determination of the coordinate position of each pixel. In Fig. 6, $F^k(p_x, p_y)$ is the pixel p predicted by $F^k(p)$. $\mathrm{DF}(p_x, p_y)$ is the position of the corresponding pixel p in the DF. $F^k(p_x, p_y)$ points to $\mathrm{DF}(p_x, p_y)$ to move gradually, and there is only one unique directional vector $\longrightarrow_{bp}$ for each point. Concretely, the improved feature map $F^k(p) \in \mathbb{R}^{C\times H\times W}$ is updated iteratively through the position pointed by $\longrightarrow_{bp}$ provided by $\mathrm{DF}(p)$, and it is calculated by bilinear interpolation. For $F^0(p)$, the corrected feature map $F^1(p) \in \mathbb{R}^{C\times H\times W}$ can be obtained after one correction process S1. After N corrections, a fine segmentation result map is obtained. The whole procedure is formalized as follows:

$$\forall p \in \Omega, \quad F^k(p) = F^{(k-1)}(p_x + \mathrm{DF}(p)_x, p_y + \mathrm{DF}(p)_y) \quad (10)$$

where $1 \le k \le N$ denotes the current step, and N is the total number of steps (set to 5 if not stated otherwise).

After performing the above rectification process, we concatenate $F^N(p)$ with $F^0(p)$ and then apply the final classifier to the concatenated feature map to predict final building segmentation.

D. Overall Loss Function

To prevent overfitting of the network model, a hybrid loss function is used to optimize the gradient drop. This study uses different loss functions for DF and segmentation results with joint supervision during training. The overall loss L can be calculated by

$$L = \alpha L_{\mathrm{BCE}} + \beta L_{\mathrm{dice}} + \gamma L_{\mathrm{DF}} \quad (11)$$

where L_{BCE} and L_{dice} are the loss functions that act on the network segmentation results. L_{DF} is the loss equation for the directional field. $\alpha, \beta,$ and γ denote the balance factors of the three loss functions. Usually, they are set to 0.5, 0.4, and 0.1, respectively.

The binary entropy loss L_{BCE} and dice loss L_{dice} are common losses of semantic segmentation and are the most appropriate for monitoring the result. Formally, L_{BCE} is defined as

$$L_{\mathrm{BCE}} = -\frac{1}{N}\sum_i (y_i \times \log(\tilde{y}_i) + (1 - y_i) \times \log(1 - \tilde{y}_i)) \quad (12)$$

where y_i and $\tilde{y}_i$ denote the true ground label probability of sample i and the predicted probability, respectively. N represents the number of pixels in the image. In addition, L_{dice} is defined as

$$L_{\mathrm{dice}} = 1 - \frac{2\sum_i^N y_i \times \tilde{y}_i + 1}{\sum_i^N y_i^2 + \sum_i^N \tilde{y}_i{}^2 + 1}. \quad (13)$$

In particular, this study uses L_{DF}, which is composed of the L_2norm distance and angle distance. L_{DF} can be expressed as

$$L_{\mathrm{DF}} = \sum_{p_i\in\Omega}\Big(\left\|\mathrm{DF}(p_i) - \tilde{\mathrm{DF}}(p_i)\right\|_2 + \alpha \times \left\|\cos^{-1}\left\langle \mathrm{DF}(p_i), \tilde{\mathrm{DF}}(p_i)\right\rangle\right\|^2\Big) \quad (14)$$

where $\mathrm{DF}(p_i)$ and $\tilde{\mathrm{DF}}(p_i)$ are the direction truth probability of sample i and the predicted DF probability, respectively. To balance the weight ratio between the L_2norm distance and angle distance, this study sets a hyperparameter α, which is set to 1 as the default value in all the experiments.

III. Experiments

A. Dataset

To evaluate the effectiveness of the method proposed in this study, we use the Massachusetts building dataset [55], the WHU aerial buidling dataset [56], and the Inria aerial image labeling dataset [57] to carry out verification experiments.

The Massachusetts building dataset consists of 151 aerial images in the Boston area, each with a size of 1500 × 1500 pixels and an area of 2.25 km^2. Therefore, the entire dataset covers approximately 340 km^2. The dataset consists of 137 training set images, with ten images in the test set and a validation set of four images. It mainly covers cities and suburbs as well as buildings of various sizes, including residential areas, commercial areas, industrial areas, etc. The spatial resolution of the dataset is 1 m.

The WHU aerial buidling dataset consists of aerial imagery and satellite imagery. This study selects aerial imagery, as this type of imagery is widely used in the current work. The aerial dataset consists of more than 220 000 independent buildings extracted from aerial images with 0.3-m ground resolution over 450 km^2 covering Christchurch, New Zealand. The dataset includes a training set of 4736 images, a verification set of 1036 images, and a test set of 2416 images. Each image and the corresponding label are cropped to 512 × 512 pixels.

The Inria aerial image labeling dataset covers an area of 810 km^2 from five cities of Austin, Chicago, Kitsap, Tyrol, and Vienna. These areas cover different urban settlements, ranging from densely populated areas to alpine towns. The dataset has 180 labeled images and 180 unlabeled images, each of size 5000 × 5000. The spatial resolution of the dataset is 0.3 m. Similar to other studies, this study uses only

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180 labeled images for training and testing. The first five images of each city are selected as the test set, and the rest are taken for training and validation sets.

B. Implementation Details

This study crops both the image data and the label data to a size of 256 × 256 pixels in the model input. The cropped images are subjected to data enhancement methods to obtain the preprocessed dataset. These data enhancement methods mainly include random flipping and rotation, addition of noise points, blurring, and gray value transformation. These methods can increase the diversity of data. After that, all the models are trained with the same parameter settings and environment. Specifically, our work is carried out on the Ubuntu 18.04 platform using the PyTorch 1.7 framework based on Python 3.7 programming language. Nvidia GTX3080 (memory 10 G) used allows a batch size of four images. This study uses the Adam optimizer to train the model. The initial learning rate is 10^{-4}. When the loss is not reduced for more than five iterations, the learning rate declines to 0.2 times the original basis. We set the weight decay to 10^{-5} to prevent overfitting. The proposed method is trained for 200 epochs on the Massachusetts building dataset, the WHU aerial building dataset, and the Inria aerial image labeling dataset and reaches a state of convergence.

C. Evaluation Metrics

To quantitatively measure the performance of BOMSC-Net, this study uses several common evaluation indicators in semantic segmentation: overall accuracy (OA), precision (P), recall (R), F_1-score, and intersection over union (IoU). Specifically, the above indicators are expressed by the following formulas:

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}} \tag{15}$$

$$P = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{16}$$

$$R = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{17}$$

$$F_1 = \frac{2 \times P \times R}{P + R} \tag{18}$$

where TP, TN, FP, and FN represent the true positives, true negatives, false positives, and false negatives, respectively.

OA is the percentage of the total sample in which prediction is correct. Generally, the higher the accuracy, the better the segmentation effect. P is the probability of the samples that are positive among all the samples that are predicted to be positive. R represents the probability of being predicted to be a positive sample for a sample that is positive. The F_1-score comprehensively considers the relationship between P and R so that the two can be simultaneously maximized and balanced. The IoU can determine the degree of the capture of the target features, maximize the intersection of the predicted label and the annotation, and determine the accuracy of the model. The specific calculation method is as follows:

$$\mathrm{IoU} = \frac{\left|P_p \cap P_t\right|}{\left|P_p \cup P_t\right|} \tag{19}$$

where P_p denotes the set of pixels predicted as buildings and P_t denotes the ground-truth set. $|\cdot|$ denotes the function for calculating the number of pixels in the set.

In addition to the above evaluation metrics, to verify the effectiveness of the method in boundary shape learning, two boundary metrics are used for quantitative evaluation: Hausdorff distance (HD) [57] and structural similarity (SSIM) [58].

As a measure of shape similarity, HD is more sensitive to segmented boundaries. The specific description of this term is as follows:

$$\begin{aligned} d_H(X, Y) &= \max\{d_{XY}, d_{YX}\} \\ &= \max\left\{\max_{x \in X} \min_{y \in Y}(x, y), \max_{y \in Y} \min_{x \in X}(x, y)\right\} \end{aligned} \tag{20}$$

where X and Y denote the ground truth and prediction map, respectively. $d_H(X, Y)$ can be understood as the maximum value of the shortest distance from a point in one point set to another point set. To eliminate the influence of a very small subset of outliers, HD is multiplied by 95% to obtain the final evaluation index (95% HD). When the distance is smaller, the shape of the predicted result is more similar to the shape of the real label.

SSIM is a measure of the similarity between two images. It is composed of a brightness function $l(X, Y)$, contrast function $c(X, Y)$, and structure function $s(X, Y)$. The specific description is as follows:

$$\begin{aligned} S(X, Y) &= F(l(X, Y), c(X, Y), s(X, Y)) \\ &= \frac{\left(2\mu_x\mu_y + C_1\right)\left(2\sigma_{xy} + C_2\right)}{\left(\mu_x^2 + \mu_y^2 + C_1\right)\left(\sigma_x^2 + \sigma_y^2 + C_2\right)} \end{aligned} \tag{21}$$

where μ, σ, and σ_{xy} represent the mean, variance, and covariance, respectively. To avoid division by zero, C_1 and C_2 represent the constants 6.50 and 58.52. The range of SSIM is $(-1, 1)$. When the two images are the same, the value of SSIM is equal to 1.

IV. Results and Discussion

A. Experiment Using the Massachusetts Building Dataset

Fig. 7 shows the experimental results of the method in this study, the classic semantic segmentation models, and the latest building extraction models. There are four rows and eight columns in Fig. 7. We choose four representative images in the test set to display, as shown in Fig. 7(a)–(d). UNet [59], DeepLabv3+ [60], SegNet [61], MAPNet [62], BMFR-Net [63], and BOMSC-Net were used for comparative experiments. The second column in Fig. 7 is the ground truth, and the following columns are the test results obtained by different models. Due to insufficient memory, we cropped the test set into overlapping patches of 1280 × 1280. Then, the patch boundary problem can be alleviated by calculating the average value of multiple predictions in the overlapping areas.

As shown in Fig. 7, the Massachusetts building dataset presents the following challenges. First, small building shape segmentation is difficult. Since the dataset has a low resolution and is dominated by small and dense buildings, the boundary segmentation of smaller buildings is often blurred. Second, there are difficulties in complete segmentation of

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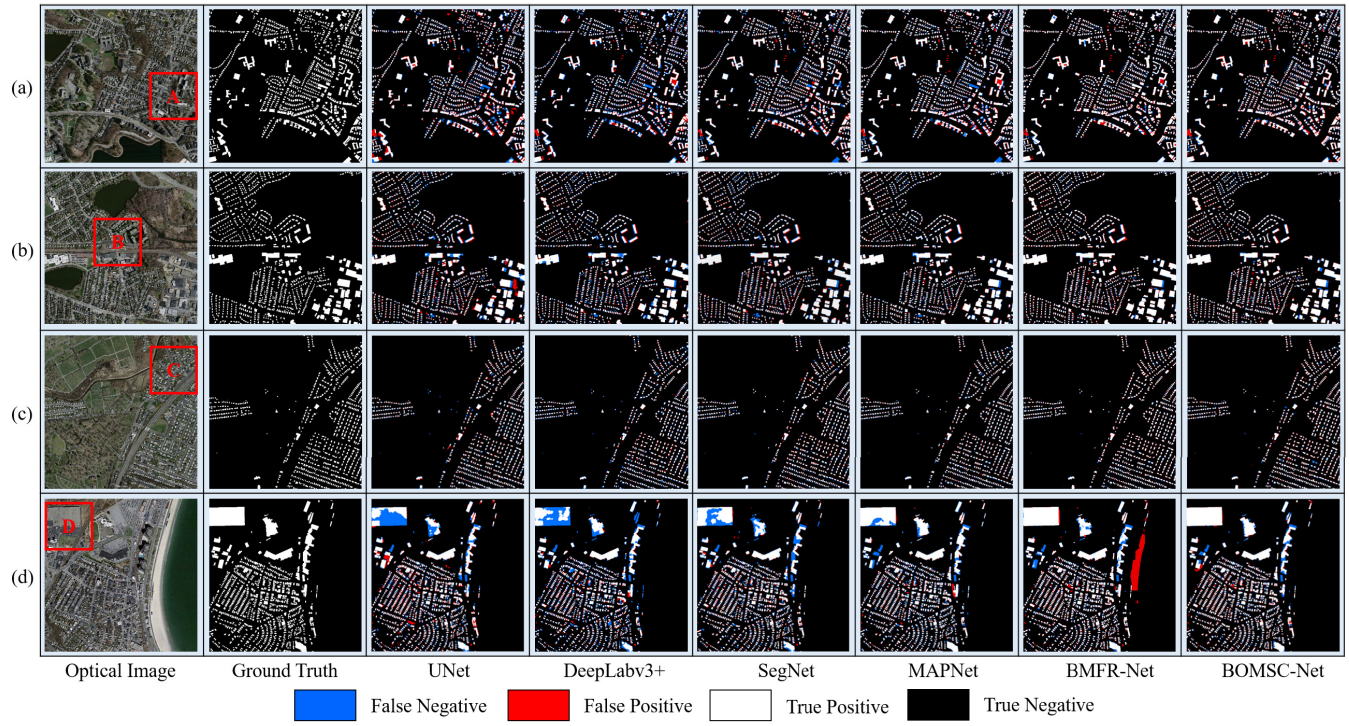

Fig. 7. Building extraction results on the Massachusetts building dataset. (a)–(d) Representative images in the test set.

large buildings. Different roof materials and partial shadow occlusion greatly reduce the integrity of building extraction. Compared with other methods, BOMSC-Net can optimize the boundaries of buildings and retain their original shapes as much as possible. Moreover, this method realizes the complete extraction of large buildings. Fig. 8 shows a detailed view of the marked area in Fig. 7.

In Fig. 8, the first and second rows contain close-ups of the extraction results of buildings in a complex environment; the third row presents the detailed results of the extraction of small buildings; and the fourth row includes the extraction results of large buildings. Due to the interference caused by shadow and tree occlusion, the other five test models experience more building omissions and misclassifications in areas A and B. In contrast, our network can better solve these problems. Regarding the problem of fuzzy boundary segmentation of small buildings in area C, it can be found from the segmentation results that BOMSC-Net can accurately extract the outlines of small buildings. Given the difficulty of segmenting large buildings in area D, BOMSC-Net can realize complete extraction of large buildings. Compared with other methods, our method offers obvious advantages.

This study uses common semantic segmentation evaluation metrics to verify the accuracy of BOMSC-Net on the Massachusetts building dataset. In addition, the evaluation metrics of the boundary are used to verify the advantages of the network in building shape learning. Since the Massachusetts building dataset is public, much work has been published in related journals. Therefore, in addition to the above comparison methods, this study quotes the accurate reports published for BRRNet [16], GMEDN [32], and DR-Net [26] in reliable journals. The results of the quantitative evaluation are shown in Table I. The rows are the evaluation metric results of each method. The columns indicate the test methods. The last row records the results for each indicator of the method proposed in this study. The best results are in bold, and the secondary results are underlined.

Table I shows that BOMSC-Net has achieved excellent results for various evaluation indicators. Specifically, the R-value and the P-value of BOMSC-Net are 83.68% and 86.64%, respectively. However, BOMSC-Net achieved the highest OA, F_1, and IoU, at 94.71%, 85.13%, and 74.71%, respectively. This shows that our method can maintain a balance between R-value and P-value, maximizing identification while reducing building misclassifications and omissions. In addition, the method in this study obtains the second lowest 95% HD score and the highest SSIM score, at 217.10% and 80.54%, respectively. The 95% HD score of DeepLabv3+ is lower than our network's 95% HD score because DeepLabv3+'s omissions are more serious and do not detect certain buildings. Moreover, the SSIM score of DeepLabv3+ is the lowest. On the whole, our network is still advanced concerning learning about the boundaries of buildings. The lower the 95% HD score, the closer the shape similarity between the building segmentation result and the ground truth. The closer the SSIM score to 1, the closer the similarity between our predicted building image and the real label.

B. Experiment Using the WHU Aerial Building Dataset

To further test the generalization performance of BOMSC-Net regarding extracting buildings in different datasets, this study also includes experiments on the WHU aerial building dataset. The WHU aerial building dataset

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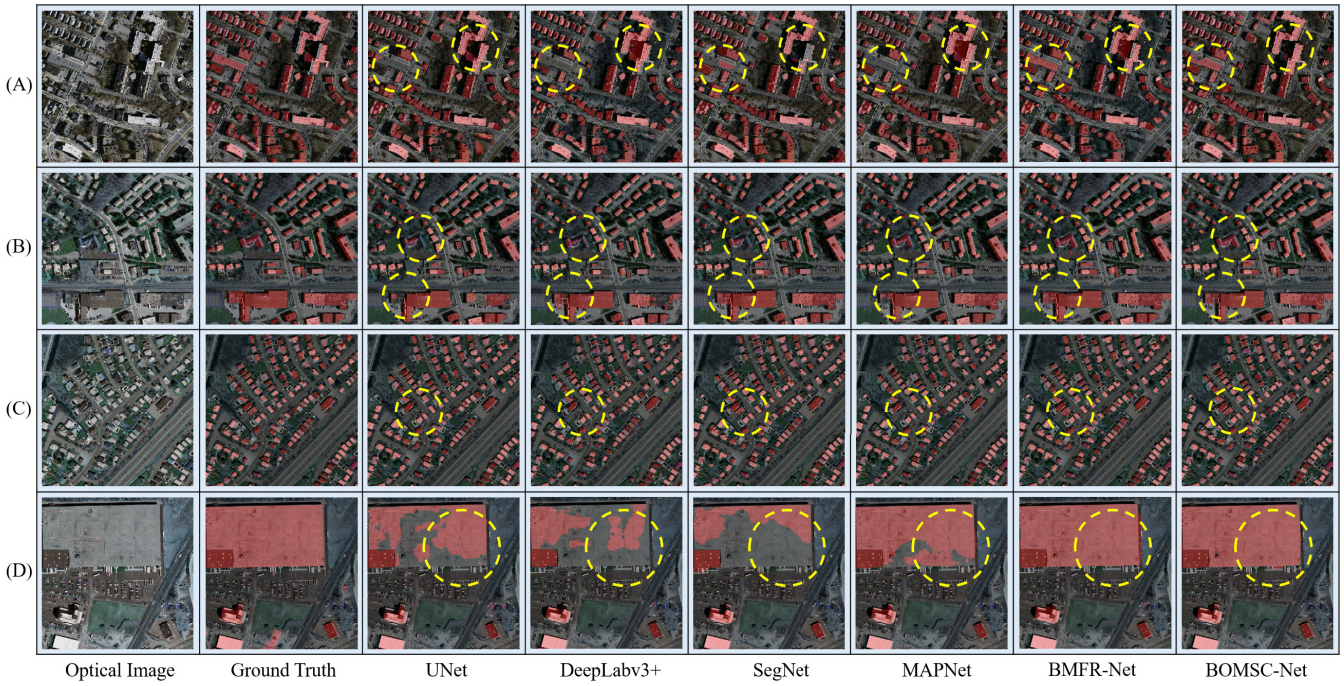

Fig. 8. (A)–(D) Images of the regions marked by red boxes in Fig. 7.

TABLE I

QUANTITATIVE EVALUATION ON THE MASSACHUSETTS BUILDING DATASET

Method	Common Metrics					Boundary Metrics	
	OA (%)	R (%)	P (%)	F_1 (%)	IoU (%)	95% HD	SSIM (%)
UNet	93.02	77.92	82.54	80.61	66.89	321.00	74.12
DeepLabv3+	92.56	73.68	83.29	78.19	64.19	**168.53**	70.94
SegNet	93.62	78.59	85.00	81.67	69.02	331.05	75.92
MAPNet	94.22	80.20	**86.84**	83.39	71.51	289.68	77.13
BMFR-Net	94.52	**85.20**	84.64	84.92	73.79	283.58	80.11
BRRNet	-	-	-	84.56	73.25	-	-
GMEDN	93.78	-	-	-	70.39	-	-
DR-Net	-	-	-	79.50	66.00	-	-
BOMSC-Net	**94.71**	83.68	86.64	**85.13**	**74.71**	217.10	**80.54**

TABLE II

QUANTITATIVE EVALUATION ON THE WHU AERIAL BUILDING DATASET

Method	Common Metrics					Boundary Metrics	
	OA (%)	R (%)	P (%)	F_1 (%)	IoU (%)	95% HD	SSIM (%)
UNet	97.27	92.52	91.86	92.19	85.51	93.11	88.83
DeepLabv3+	97.37	91.27	93.45	92.35	85.78	97.58	89.03
SegNet	97.41	92.35	92.73	92.54	86.12	97.63	89.33
MAPNet	98.17	93.84	**95.59**	94.70	89.94	84.24	92.36
BMFR-Net	97.92	92.86	95.06	93.95	88.59	90.13	91.06
SRI-Net	-	93.28	95.21	94.23	89.09	-	-
DR-Net	-	-	-	93.80	88.30	-	-
SST(*R*18,*S*4)	**98.80**	-	-	94.13	89.01	-	-
BOMSC-Net	98.20	**94.50**	95.14	**94.80**	**90.15**	**80.03**	**92.43**

distribution is equally complex, including buildings with different materials, sizes, and shapes. The prediction results based on the WHU aerial building dataset are shown in Fig. 9. The columns display the results of the test methods, and the rows represent samples of different buildings.

Fig. 9 shows that the proposed BOMSC-Net offers more advantages than the other test methods when dealing with buildings in complex scenarios. The buildings extracted by BOMSC-Net are the most complete, and its boundary shape segmentation is better. The first and second rows show

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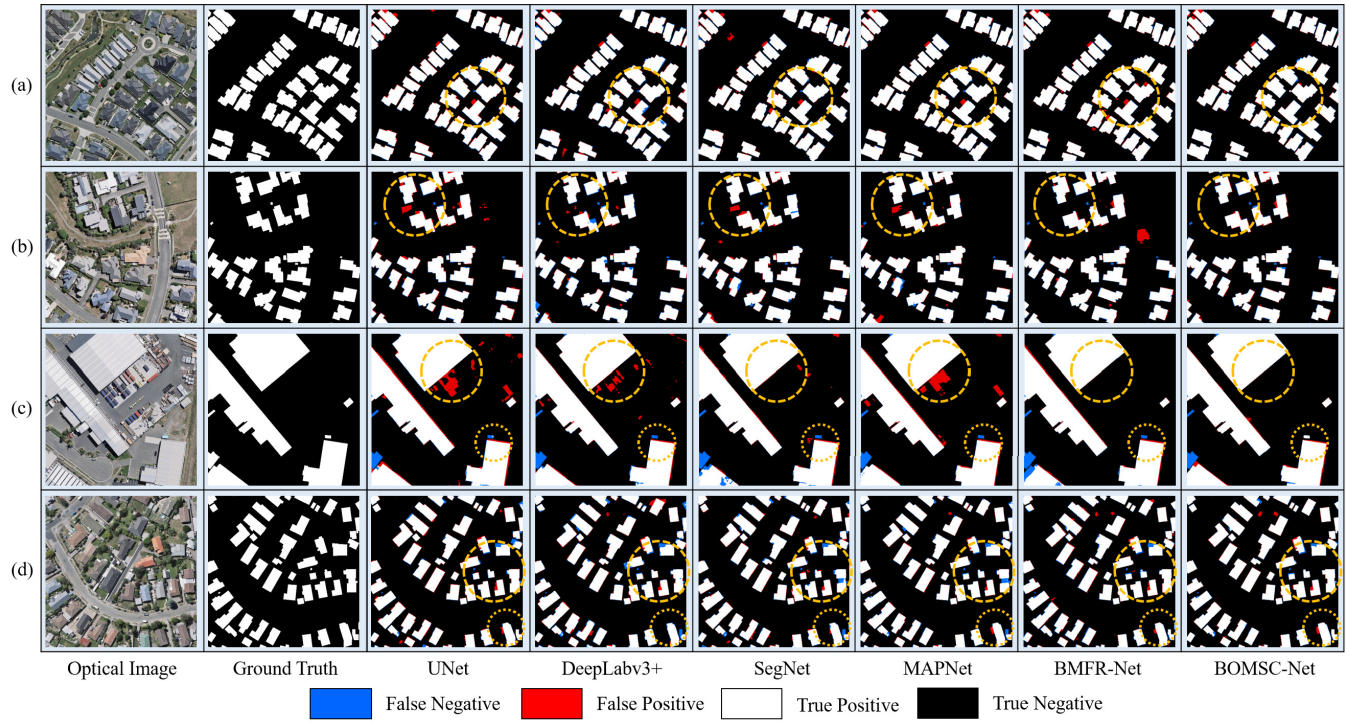

Fig. 9. Building extraction results on the WHU aerial building dataset. (a)–(d) Representative images in the test set.

buildings with complex boundary contours. The method proposed in this study is more accurate and detailed regarding the description of the shape of the buildings. BOMSC-Net can accurately distinguish the building and the background for incorrectly marked buildings. As shown in the second and third rows in Fig. 9, BOMSC-Net can significantly reduce misclassification and omission for other objects with obvious interference. When the roof material of the building changes, the segmentation result is incomplete. This study can fully perceive the contextual information of the building with the advantage of the GRB and accurately infer the complete shape of the building.

The results of the quantitative evaluation are shown in Table II. The rows represent different test methods, and the columns represent different evaluation indicators. In addition to the above test models, this study cited the latest building extraction studies SRI-Net [64], DR-Net [26], and SST [65] to further compare the building segmentation results. BOMSC-Net obtained the highest R-value, F_1, and IoU, at 98.20%, 94.50%, 94.80%, and 90.15%, respectively. This further verifies that our method can ensure the integrity of the building even when the roof material is complex and trees are severely impeded. There are various containers around the factory building in the WHU aerial building dataset. These containers often have spectrum, texture, shape, and color characteristics similar to those of factory buildings, as shown in the third row in Fig. 9, resulting in slightly lower OA and P-values for our method than other methods, at 95.14% and 98.20%, respectively. BOMSC-Net obtains the lowest 95% HD and highest SSIM scores on the WHU aerial building dataset, with 80.03% and 92.43%, respectively. It is sufficient to prove that the method proposed in this study has advantages in shape learning. The DFOM can optimize the boundaries of buildings, achieving fine segmentation of buildings.

C. Experiment Using the Inria Aerial Image Labeling Dataset

Since the Inria Aerial Image Labeling Dataset contains aerial images of buildings in five different cities, it is very convincing to test the generalization of the model. Therefore, this study also conducts comparative experiments on this dataset. As shown in Fig. 10(a), (b), (c), (d), and (e), the rows show the test images of different cities: Austin (U.S.), Chicago (U.S.), Kitsap (U.S.), Tyrol (Austria), and Vienna (Austria)., respectively The columns show the test results of different models. Likewise, the test set is overlapped and cropped due to insufficient device memory. The size of each image is 1280 × 1280, for a total of 400 images. As can be seen from Fig. 10, even in complex scenes that include tree occlusion, mislabeling, or buildings of different sizes and styles, BOMSC-Net can still extract building results with more complete boundary shapes compared with the results from other models. Fig. 10(a) and (b) shows the segmentation results of buildings with significant size differences. Obviously, BOMSC-Net can perceive long-distance context information, which can help complete the segmentation of large-scale buildings. Fig. 10(c) and (d) shows the extraction results of small buildings by different models. It can be seen from the highlighted positions in Fig. 10 that other models have certain misclassifications and omissions. BOMSC-Net can perfectly solve this problem. Fig. 10(e) shows a mislabeled image. In complex scenes, the blurring of background and building boundaries creates severe difficulties in accurately extracting buildings. However, it can be seen from Fig. 10(e) that com-

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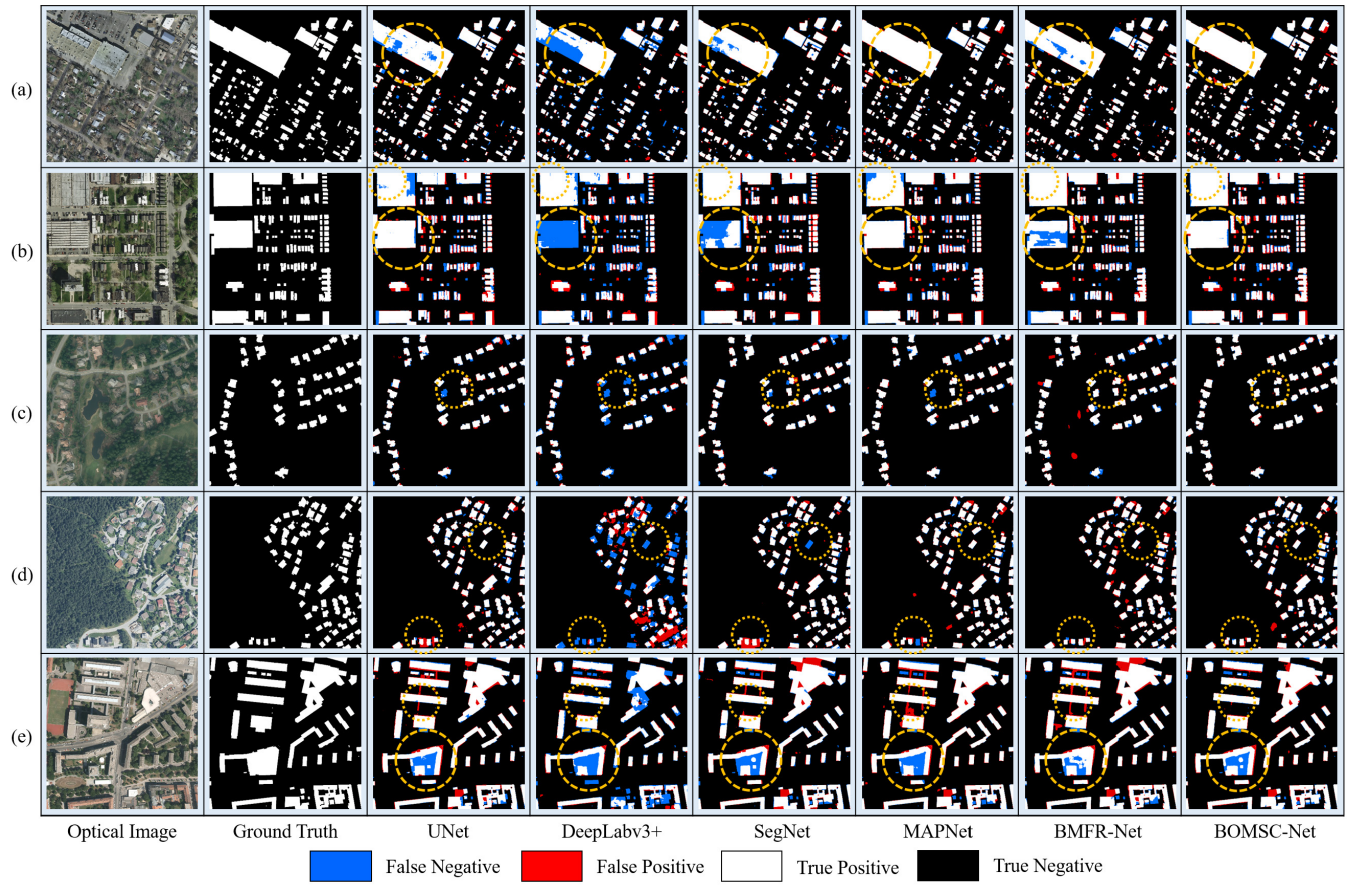

Fig. 10. Building extraction results on the Inria Aerial Image Labeling Dataset. Rows show test images for different cities. (a) Austin (USA). (b) Chicago (USA). (c) Kitsap (USA). (d) Tyrol (Austria). (e) Vienna (Austria).

pared with other models, BOMSC-Net can accurately identify the shape of buildings and achieve refined segmentation.

In this study, a quantitative evaluation of the Inria Aerial Image Labeling Dataset was carried out, and the evaluation results are shown in Table III. BOMSC-Net achieves the highest scores in terms of F_1, IoU, R-value, and P-value. Therefore, comprehensive evaluation of the method proposed in this article is better than other methods, and BOMSC-Net is more robust and generalizable. Specifically, the R-value is significantly higher than that of other methods, 1.57% higher than the second place. This shows that BOMSC-Net can effectively solve the missing segmentation of buildings and achieve complete results in the face of large-scale buildings. Boundary optimization and integrity descriptions of buildings are complementary. Therefore, effective boundary optimization can better promote the improvement of regional accuracy. In particular, in the boundary evaluation metrics, BOMSC-Net achieves the highest scoring SSIM and the second lowest scoring 95% HD, which are 84.31% and 285.79, respectively. The proposed BOMSC-Net can be more advantageous in learning boundary features.

D. Ablation Analyses

1) Influence of Different Modules: This study included ablation experiments on the Massachusetts building dataset to better understand the impact of each module in the proposed BOMSC-Net. The basic network, the MSCAM, and the DFOM are deleted or replaced in BOMSC-Net to show the influence of these modules on the network.

First, this study verifies the effectiveness of the basic network of BOMSC-Net. Second, we remove MSCAM, leaving only the basic network and DFOM, to verify the effectiveness of DFOM in BOMSC-Net. Third, we remove DFOM and add MSCAM based on the basic network. Since DFOM is removed, we adjust the loss function. This study removes L_{DF} from the hybrid loss function, leaving only L_{BCE} and L_{dice}. Simultaneously, α and γ are set to 0.6 and 0.4, respectively. Second, we add MSCAM based on the basic network, and the loss remains the same. Finally, we stack DFOM based on this network, and L_{DF} needs to be introduced in the hybrid loss function. At this time, the hybrid loss function is mentioned above. Fig. 11 shows some representative experimental results. Fig. 11(A) and (B) shows optical images and the ground truth, respectively. Fig. 11(C)–(F) shows the test results of the basic network, basic network with DFOM, basic network with MSCAM, and BOMSC-Net, respectively.

Fig. 11 shows that when the basic network is combined with different modules, the results of building segmentation in different complex scenes are changed. This study shows the segmentation results in Fig. 11(a), (b), (c), (d), and (e)—roof shadows, incorrect labels, material changes, tree occlusions, and complex building shapes, respectively. These are shown

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TABLE III
QUANTITATIVE EVALUATION ON THE INRIA AERIAL IMAGE LABELING DATASET

Method	Common Metrics					Boundary Metrics	
	OA (%)	R (%)	P (%)	F_1 (%)	IoU (%)	95% HD	SSIM (%)
UNet	94.80	84.28	86.39	85.32	74.40	319.20	79.51
DeepLabv3+	92.68	76.55	87.33	81.59	68.90	316.92	75.13
SegNet	94.66	83.83	86.03	84.91	73.78	319.67	78.95
MAPNet	95.16	85.75	87.19	86.46	76.15	**280.60**	81.54
BMFR-Net	94.71	86.01	85.52	85.52	74.71	287.92	79.63
SRI-Net	-	81.46	85.77	83.56	71.76	-	-
GMEDN	**96.43**	-	-	-	76.69	-	-
SST(*R*18,*S*4)	96.19	-	-	86.49	77.12	-	-
BOMSC-Net	95.31	**87.58**	**87.93**	**87.75**	**78.18**	285.79	**84.31**

Fig. 11. Building extraction results of different module combinations. Segmentation is shown with (a) roof shadows, (b) incorrect labels, (c) material changes, (d) tree occlusions, and (e) complex building shapes. The results are shown in (A) optical images, (B) ground truth, (C) basic network, (D) basic network + DFOM, (E) basic network + MSCAM, and (F) BOMSC-Net, respectively.

in Fig. 11(a)–(e). The results of the third column [Fig. 11(C)] show that the basic network already has good learning capabilities and can accurately segment buildings under these complex scenarios. However, it is clear that the segmentation results still need to be improved in terms of shape and building integrity. [Fig. 11(D)] shows the effect of adding DFOM to the basic network. Fig. 11 shows that the extracted building boundary has been significantly improved. As shown in the fifth column [Fig. 11(E)], the basic network with the MSCAM can effectively infer the contextual information of the buildings in scenes with shadows, occlusions, etc., to learn complete building shape. Finally, the sixth column shows the visible view with BOMSC-Net. The shape of the building is better optimized than in the previous results, and its edges are more regular.

In addition, we performed a quantitative evaluation, and the results are shown in Table IV. Table IV shows that our basic network can obtain higher accuracy and scores than the commonly used semantic segmentation networks UNet, SegNet, and DeepLabv3+. The network performance

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TABLE IV
QUANTITATIVE EVALUATION OF ABLATION EXPERIMENTS

Method	Common Metrics					Boundary Metrics	
	OA (%)	R (%)	P (%)	F_1 (%)	IoU (%)	95% HD	SSIM (%)
Basic network	93.94	78.41	**86.83**	82.41	70.08	312.74	76.42
+DFOM	94.28 (+0.34)	83.19 (+4.78)	84.89 (-1.94)	84.03 (+1.62)	72.46 (+2.38)	244.40 (-68.34)	78.87 (+2.45)
+ MSCAM	94.34 (+0.40)	81.64 (+3.23)	86.36 (-0.47)	83.94 (+1.53)	72.32 (+2.24)	268.75 (-43.99)	77.84 (+1.42)
BOMSC-Net	**94.71** (+0.77)	**83.68** (+5.27)	86.64 (-0.19)	**85.13** (+2.72)	**74.71** (+4.63)	**217.10** (-95.64)	**80.54** (+4.12)

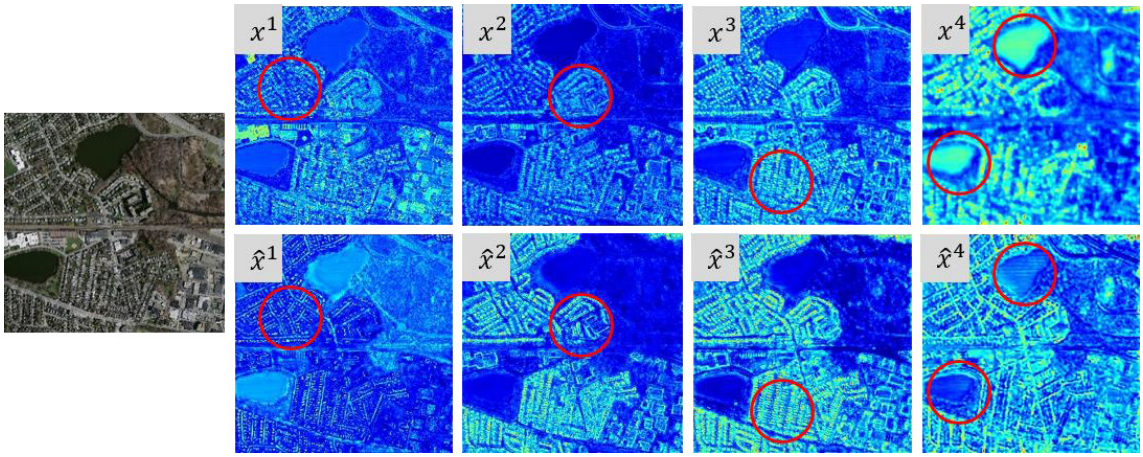

Fig. 12. Visualization of feature maps for AGs. x^l and $\hat{x}^l, l \in [1, 2, 3, 4]$ represent the input and output feature maps of each layer of AGs.

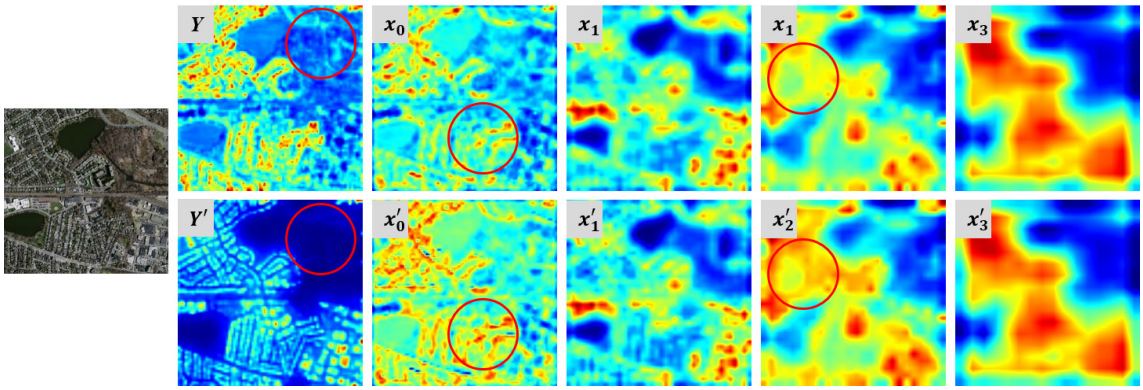

Fig. 13. Feature map visualization for GRBs. Y and Y' represent the input and output feature maps of MSCAM, respectively. x_i and $x'_i, i \in [0, 1, 2, 3]$ represent the input and output feature maps of GRBs, respectively. Among them, x_0 represents the feature map without max-pooling operation; x_1, x_2, x_3 represent the feature maps with max-pooling operations whose kernel size is 2×2, 3×3, and 5×5.

TABLE V
INFLUENCE OF AG

Method	Common Metrics					Boundary Metrics	
	OA (%)	R (%)	P (%)	F_1 (%)	IoU (%)	95% HD	SSIM (%)
Basic network without AGs	93.02	77.92	82.54	80.61	66.89	321.00	74.12
Basic network	93.94 (+0.92)	78.41 (+0.49)	86.83 (+4.29)	82.41 (+1.80)	70.08 (+3.19)	312.74 (-8.26)	76.42 (+2.30)

improves between the R-value and P-value, which reduces building omissions and misclassifications as much as possible. The network can obtain a higher score at the cost of reducing the accuracy. Compared with the results of the basic network, most indicators increased significantly, among which the R-value increased by 4.78%. This indicates that the DFOM rectifies the direction through DF, which is helpful for the network to learn correct boundary. However, the P-value of the "basic network + DFOM" decreased by 1.94%. This implies that the DFOM cannot guarantee the recognition of more

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TABLE VI
INFLUENCE OF GRAPH REASONING BLOCK

Method	Common Metrics					Boundary Metrics	
	OA (%)	R (%)	P (%)	F_1 (%)	IoU (%)	95% HD	SSIM (%)
Basic network	93.94	78.41	**86.83**	82.41	70.08	312.74	76.42
+ MSCAM without GRBs	94.02 (+0.08)	80.30 (+1.89)	85.74 (-1.09)	82.93 (+0.52)	70.84 (+0.76)	280.48 (-32.26)	77.57 (+1.15)
+ MSCAM	94.34 (+0.40)	81.64 (+3.23)	86.36 (-0.47)	83.94 (+1.53)	72.32 (+2.24)	268.75 (-43.99)	77.84 (+1.42)

TABLE VII
INFLUENCE OF RECTIFICATION STEPS

rectification step	Common Metrics					Boundary Metrics	
	OA (%)	R (%)	P (%)	F_1 (%)	IoU (%)	95% HD	SSIM (%)
N=1	94.26	82.14	**85.59**	83.83	72.16	260.87	78.61
N=3	94.22	82.98	84.78	83.87	72.22	**229.85**	78.70
N=5	**94.28**	83.19	84.89	**84.03**	**72.46**	244.40	**78.87**
N=7	94.18	83.31	84.35	83.83	72.16	293.50	78.48
N=9	94.12	**83.76**	83.74	83.75	72.04	388.46	78.49

complete buildings. At the same time, compared with the basic network, the P-value of "basic network + MSCAM" only decreased by 0.47%, while the others increased significantly. This demonstrates that the MSCAM can effectively alleviate the incompleteness of building extraction and achieve accurate segmentation as much as possible. Ultimately, the P-value of BOMSC-Net was only slightly reduced by 0.19%, while the other indicators were improved. OA, R-value, F_1, and IoU increased by 0.77%, 5.27%, 2.72%, and 4.63%, respectively. This shows that the common use of DFOM and MSCAM can promote the effectiveness of the network. In addition to the great improvement in the semantic segmentation metrics, BOMSC-Net makes progress in the boundary metrics. After adding DFOM to the basic network, 95% HD and SSIM are particularly improved. The 95% HD decreases by 68.34, while SSIM increases by 2.45%. This shows that DFOM significantly promotes the shape description of buildings. Table IV shows that the basic network with MSCAM can also improve the network performance as much as possible and optimize building boundaries. In the end, compared with the basic network, 95% HD of BOMSC-Net was significantly reduced by 95.64, and SSIM was also significantly increased by 4.12%. This shows that BOMSC-Net can realize optimization and complete description of the boundary information of the building.

2) Influence of the AGs: To verify the effectiveness of AGs, this study conducted experiments based on the basic network. Specifically, all the AGs in the basic network are removed, and only the backbone is retained. At this time, the loss function is consistent with the ablation experiment of the basic network. The results of the accuracy evaluation are shown in Table V. It can be seen from the table that the P-value is significantly improved in the result of adding AGs. More buildings can be accurately identified, which indicates that AGs can encourage the model to focus on salient building areas in the images. In Fig. 12, this study visualizes the input feature map x^l and output feature map $\hat{x}^l, l \in [1, 2, 3, 4]$ of each layer of AGs. Fig. 12 shows that $\hat{x}^l$ pays more attention to buildings than x^l. In particular, $\hat{x}^1$ is more prominent than low-level characteristics of x^1, but the semantic information is not rich enough; $\hat{x}^4$ has more complete semantic information than x^4, especially $\hat{x}^4$ which has a significantly lower impact on the water body than x^4.

3) Influence of the GRBs: To analyze the influence of GRBs, this study conducted experiments based on "basic network + MSCAM." Specifically, this study removed all the GRBs in the MSCAM and retained only other convolution operations and maximum pooling operations. Then, we compare the experimental results of "basic network + MSCAM" and "basic network + MSCAM without GRBs" on the Massachusetts building dataset. As shown in Table VI, GRBs have advantages over all the indicators without GRBs. GRBs can significantly increase the R-value and minimize the loss of the P-value, which shows that GRBs help learn complete building information. The results of visualization are shown in Fig. 13. Y and Y' represent the input and output feature maps of the MSCAM, respectively. It can be clearly seen that Y' can effectively perceive the global information of the building. In particular, the irrelevant background in Y' can be effectively eliminated. x_i and $x_i', i \in [0, 1, 2, 3]$ represent the input and output feature maps of GRBs, respectively, where x_0 represents the feature map without max-pooling operation, and x_1, x_2, x_3 represents the feature maps with max-pooling operations whose kernel size is 2×2, 3×3, and 5×5.

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TABLE VIII
QUANTITATIVE EVALUATION OF PARAMETERS OF DIFFERENT MAIN MODULES

	basic network	basic network + DFOM	basic network + MSCAM	BOMSC-Net
Params(M)	34.879	34.887 (+0.008)	129.309 (+94.43)	129.317 (+94.438)

Obviously, x_0' and x_2' can highlight the building area more than x_0 and x_2.

4) Influence of the Rectification Step (N): In this study, experiments are conducted based on the "basic network + DFOM" to explore the influence of rectification N on the network. Set $N = 1$, 3, 5, 7, and 9, and the experimental results are shown in Table VII. Table VII shows that OA, F_1, and IoU are basically in a normal distribution. Taken together, when $N = 5$, OA, F_1, IoU, and SSIM all achieved the highest scores. The feature optimization of DFOM cannot be satisfied when N is too small; the rectification result deviates from the true value when N is too large.

5) Parameter Analysis: This study also counted the model parameters of different main modules. Specifically, this study calculated the model parameters of the basic network, "basic network + DFOM," "basic network + MSCAM," and BOMSC-Net. It can be seen from Table VIII that the number of model parameters of "basic network + DFOM" has only increased by 0.008(M) compared with the basic network. On the other hand, compared with the basic network, the model parameter amount of "basic network + MSCAM" has increased by 94.43(M). Compared with MSCAM, DFOM has only a slight increase in the parameters of BOMSC-Net. Because four GRBs are used in MSCAM, the parameter quantity of the model is obviously increased.

V. CONCLUSION

This study explores the shortcomings of the existing methods for extracting buildings from high-resolution remote sensing images based on deep learning. Inaccurate boundary extraction and incomplete building extraction are the key problems of current building extraction. Therefore, this study proposes a building extraction method (BOMSC-Net) that simultaneously considers boundary optimization and multi-scale context awareness to address inaccurate building boundaries and incomplete extraction results. The experimental results on three open-source datasets show that the proposed BOMSC-Net can optimize boundaries and enhance context awareness. In complex environments, such as those with shadow and tree occlusions, BOMSC-Net can accurately extract the boundaries of buildings and can achieve complete extraction of buildings of different scales. To verify the performance of our network, this study not only uses general semantic segmentation evaluation metrics but also uses boundary evaluation metrics to quantitatively illustrate the advantages of BOMSC-Net in boundary optimization. Compared with previous work, the extraction results are significantly improved. The test results include higher R-value, F_1-score, and IoU than the comparison method, indicating that our method can maintain a balance between R-value and P-value and reduce false and missed segmentation. BOMSC-Net achieves excellent scores for 95% HD and SSIM, showing that it can achieve building boundary optimization. This study also includes ablation experiments to verify that MSCAM and DFOM significantly improve the network performance.

However, the method in this study has some shortcomings. BOMSC-Net may have the same enhancement effect on features similar to buildings, so there would be some misclassifications in places where buildings and similar features are present. The method of GRB would be restricted in cases of severe occlusion by shadows and trees. We will consider multi-spectral and multi-source data fusion to solve these problems in future work.

REFERENCES

[1] *Revision of World Urbanization Prospects*, United Nations, New York, NY, USA, 2018.
[2] Z. Huang, G. Cheng, H. Wang, H. Li, L. Shi, and C. Pan, "Building extraction from multi-source remote sensing images via deep deconvolution neural networks," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2016, pp. 1835–1838.
[3] W. Li, C. He, J. Fang, and H. Fu, "Semantic segmentation based building extraction method using multi-source GIS map datasets and satellite imagery," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2018, pp. 238–241.
[4] M. Turker and D. Koc-San, "Building extraction from high-resolution optical spaceborne images using the integration of support vector machine (SVM) classification, Hough transformation and perceptual grouping," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 34, pp. 58–69, Feb. 2015.
[5] S. A. N. Gilani, M. Awrangjeb, and G. Lu, "An automatic building extraction and regularisation technique using LiDAR point cloud data and orthoimage," *Remote Sens.*, vol. 8, no. 3, p. 258, Mar. 2016.
[6] C. Liu, X. Huang, Z. Zhu, H. Chen, X. Tang, and J. Gong, "Automatic extraction of built-up area from ZY3 multi-view satellite imagery: Analysis of 45 global cities," *Remote Sens. Environ.*, vol. 226, pp. 51–73, Jun. 2019.
[7] J. P. Cohen, W. Ding, C. Kuhlman, A. Chen, and L. Di, "Rapid building detection using machine learning," *Int. J. Speech Technol.*, vol. 45, no. 2, pp. 443–457, Sep. 2016.
[8] L. Xu, M. Kong, and B. Pan, "Building extraction by stroke width transform from satellite imagery," in *Proc. CCF Chin. Conf. Comput. Vis.*, 2017, pp. 340–351.
[9] J. Wang, X. Yang, X. Qin, X. Ye, and Q. Qin, "An efficient approach for automatic rectangular building extraction from very high resolution optical satellite imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 3, pp. 487–491, Mar. 2015.
[10] G. Liasis and S. Stavrou, "Building extraction in satellite images using active contours and colour features," *Int. J. Remote Sens.*, vol. 37, no. 5, pp. 1127–1153, Mar. 2016.
[11] L. Wang, G. Chen, D. Shi, Y. Chang, S. Chan, J. Pu, and X. Yang, "Active contours driven by edge entropy fitting energy for image segmentation," *Signal Process.*, vol. 149, pp. 27–35, Aug. 2018.
[12] D. Chai, "A probabilistic framework for building extraction from airborne color image and DSM," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 3, pp. 948–959, Mar. 2017.

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[13] M. Teimouri, M. Mokhtarzade, and M. J. Valadan Zoej, "Optimal fusion of optical and SAR high-resolution images for semiautomatic building detection," *GISci. Remote Sens.*, vol. 53, no. 1, pp. 45–62, Jan. 2016.

[14] Z. Guo and S. Du, "Mining parameter information for building extraction and change detection with very high-resolution imagery and GIS data," *GISci. Remote Sens.*, vol. 54, no. 1, pp. 38–63, Jan. 2017.

[15] L. Zhang, J. Wu, Y. Fan, H. Gao, and Y. Shao, "An efficient building extraction method from high spatial resolution remote sensing images based on improved mask R-CNN," *Sensors*, vol. 20, no. 5, p. 1465, Mar. 2020.

[16] Z. Shao, P. Tang, Z. Wang, N. Saleem, S. Yam, and C. Sommai, "BRRNet: A fully convolutional neural network for automatic building extraction from high-resolution remote sensing images," *Remote Sens.*, vol. 12, no. 6, p. 1050, Mar. 2020.

[17] H. Guo, Q. Shi, A. Marinoni, B. Du, and L. Zhang, "Deep building footprint update network: A semi-supervised method for updating existing building footprint from bi-temporal remote sensing images," *Remote Sens. Environ.*, vol. 264, Oct. 2021, Art. no. 112589.

[18] Q. Shi, M. Liu, S. Li, X. Liu, F. Wang, and L. Zhang, "A deeply supervised attention metric-based network and an open aerial image dataset for remote sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2022.

[19] S. Liu, Q. Shi, and L. Zhang, "Few-shot hyperspectral image classification with unknown classes using multitask deep learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5085–5102, Jun. 2021.

[20] Q. Shi, X. Tang, T. Yang, R. Liu, and L. Zhang, "Hyperspectral image denoising using a 3-D attention denoising network," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 12, pp. 10348–10363, Dec. 2021.

[21] C. Zhang, I. Sargent, X. Pan, H. Li, A. Gardiner, J. Hare, and P. M. Atkinson, "Joint deep learning for land cover and land use classification," *Remote Sens. Environ.*, vol. 221, pp. 173–187, Feb. 2019.

[22] D. He, Q. Shi, X. Liu, Y. Zhong, and X. Zhang, "Deep subpixel mapping based on semantic information modulated network for urban land use mapping," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 12, pp. 10628–10646, Dec. 2021.

[23] D. Zhang *et al.*, "A generalized approach based on convolutional neural networks for large area cropland mapping at very high resolution," *Remote Sens. Environ.*, vol. 247, Sep. 2020, Art. no. 111912.

[24] M. Guo, H. Liu, Y. Xu, and Y. Huang, "Building extraction based on U-Net with an attention block and multiple losses," *Remote Sens.*, vol. 12, no. 9, p. 1400, Apr. 2020.

[25] W. Boonpook, Y. Tan, and B. Xu, "Deep learning-based multi-feature semantic segmentation in building extraction from images of UAV photogrammetry," *Int. J. Remote Sens.*, vol. 42, no. 1, pp. 1–19, Jan. 2021.

[26] M. Chen *et al.*, "DR-Net: An improved network for building extraction from high resolution remote sensing image," *Remote Sens.*, vol. 13, no. 2, p. 294, Jan. 2021.

[27] S. Wang, X. Hou, and X. Zhao, "Automatic building extraction from high-resolution aerial imagery via fully convolutional encoder-decoder network with non-local block," *IEEE Access*, vol. 8, pp. 7313–7322, 2020.

[28] Z. Zhang, W. Guo, M. Li, and W. Yu, "GIS-supervised building extraction with label noise-adaptive fully convolutional neural network," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 12, pp. 2135–2139, Dec. 2020.

[29] B. Sariturk, B. Bayram, Z. Duran, and D. Z. Seker, "Feature extraction from satellite images using segnet and fully convolutional networks (FCN)," *Int. J. Eng. Geosci.*, pp. 138–143, Oct. 2020.

[30] S. He and W. Jiang, "Boundary-assisted learning for building extraction from optical remote sensing imagery," *Remote Sens.*, vol. 13, no. 4, p. 760, Feb. 2021.

[31] H. Jung, H.-S. Choi, and M. Kang, "Boundary enhancement semantic segmentation for building extraction from remote sensed image," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–12, 2022.

[32] J. Ma, L. Wu, X. Tang, F. Liu, X. Zhang, and L. Jiao, "Building extraction of aerial images by a global and multi-scale encoder-decoder network," *Remote Sens.*, vol. 12, no. 15, p. 2350, Jul. 2020.

[33] Y. Xu, L. Wu, Z. Xie, and Z. Chen, "Building extraction in very high resolution remote sensing imagery using deep learning and guided filters," *Remote Sens.*, vol. 10, no. 1, p. 144, 2018.

[34] Y. Xie *et al.*, "Refined extraction of building outlines from high-resolution remote sensing imagery based on a multifeature convolutional neural network and morphological filtering," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 1842–1855, 2020.

[35] B. Zhang, C. Wang, Y. Shen, and Y. Liu, "Fully connected conditional random fields for high-resolution remote sensing land use/land cover classification with convolutional neural networks," *Remote Sens.*, vol. 10, no. 12, p. 1889, Nov. 2018.

[36] Q. Zhu, Z. Li, Y. Zhang, and Q. Guan, "Building extraction from high spatial resolution remote sensing images via multiscale-aware and segmentation-prior conditional random fields," *Remote Sens.*, vol. 12, no. 23, p. 3983, Dec. 2020.

[37] J. Sun, W. Li, Y. Zhang, and W. Gong, "Building segmentation of remote sensing images using deep neural networks and domain transform CRF," *Proc. SPIE*, vol. 11155, Oct. 2019, Art. no. 111550N.

[38] G. Wu *et al.*, "A boundary regulated network for accurate roof segmentation and outline extraction," *Remote Sens.*, vol. 10, no. 8, p. 1195, Jul. 2018.

[39] J. Zhou, M. Hao, D. Zhang, P. Zou, and W. Zhang, "Fusion PSPnet image segmentation based method for multi-focus image fusion," *IEEE Photon. J.*, vol. 11, no. 6, pp. 1–12, Dec. 2019.

[40] S. C. Yurtkulu, Y. H. Sahin, and G. Unal, "Semantic segmentation with extended DeepLabv3 architecture," in *Proc. 27th Signal Process. Commun. Appl. Conf. (SIU)*, Apr. 2019, pp. 1–4.

[41] J. Zhang, S. Lin, L. Ding, and L. Bruzzone, "Multi-scale context aggregation for semantic segmentation of remote sensing images," *Remote Sens.*, vol. 12, no. 4, p. 701, Feb. 2020.

[42] G. Sun, H. Huang, A. Zhang, F. Li, H. Zhao, and H. Fu, "Fusion of multiscale convolutional neural networks for building extraction in very high-resolution images," *Remote Sens.*, vol. 11, no. 3, p. 227, 2019.

[43] Y. Liu, L. Gross, Z. Li, X. Li, X. Fan, and W. Qi, "Automatic building extraction on high-resolution remote sensing imagery using deep convolutional encoder-decoder with spatial pyramid pooling," *IEEE Access*, vol. 7, pp. 128774–128786, 2019.

[44] Y. Jin, W. Xu, C. Zhang, X. Luo, and H. Jia, "Boundary-aware refined network for automatic building extraction in very high-resolution urban aerial images," *Remote Sens.*, vol. 13, no. 4, p. 692, Feb. 2021.

[45] D.-Y. Chen, L. Peng, W.-C. Li, and Y.-D. Wang, "Building extraction and number statistics in WUI areas based on UNet structure and ensemble learning," *Remote Sens.*, vol. 13, no. 6, p. 1172, Mar. 2021.

[46] O. Oktay *et al.*, "Attention U-Net: Learning where to look for the pancreas," 2018, *arXiv:1804.03999*.

[47] H. Fukui, T. Hirakawa, T. Yamashita, and H. Fujiyoshi, "Attention branch network: Learning of attention mechanism for visual explanation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 10705–10714.

[48] H. Yang, P. Wu, X. Yao, Y. Wu, B. Wang, and Y. Xu, "Building extraction in very high resolution imagery by dense-attention networks," *Remote Sens.*, vol. 10, no. 11, p. 1768, 2018.

[49] X. Pan *et al.*, "Building extraction from high-resolution aerial imagery using a generative adversarial network with spatial and channel attention mechanisms," *Remote Sens.*, vol. 11, no. 8, p. 917, Apr. 2019.

[50] Y. Chen, M. Rohrbach, Z. Yan, Y. Shuicheng, J. Feng, and Y. Kalantidis, "Graph-based global reasoning networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 433–442.

[51] J. Li *et al.*, "Multi-scale GCN-assisted two-stage network for joint segmentation of retinal layers and discs in peripapillary OCT images," *Biomed. Opt. Exp.*, vol. 12, no. 4, pp. 2204–2220, 2021.

[52] F. Cheng *et al.*, "Learning directional feature maps for cardiac MRI segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, 2020, pp. 108–117.

[53] Z. Dong, J. Li, T. Fang, and X. Shao, "Lightweight boundary refinement module based on point supervision for semantic segmentation," *Image Vis. Comput.*, vol. 110, Jun. 2021, Art. no. 104169.

[54] H. Embrechts and D. Roose, "A parallel Euclidean distance transformation algorithm," *Comput. Vis. Image Understand.*, vol. 63, no. 1, pp. 15–26, Jan. 1996.

[55] V. Mnih, *Machine Learning for Aerial Image Labeling*. Toronto, ON, Canada: Univ. Toronto, 2013.

[56] S. Ji, S. Wei, and M. Lu, "Fully convolutional networks for multisource building extraction from an open aerial and satellite imagery data set," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 574–586, Jan. 2019.

[57] D. P. Huttenlocher, G. A. Klanderman, and W. J. Rucklidge, "Comparing images using the Hausdorff distance," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 15, no. 9, pp. 850–863, Sep. 1993.

[58] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Trans. Image Process.*, vol. 13, no. 4, pp. 600–612, Apr. 2004.

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[59] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, 2015, pp. 234–241.
[60] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder–decoder with atrous separable convolution for semantic image segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 801–818.
[61] V. Badrinarayanan, A. Kendall, and R. Cipolla, "SegNet: A deep convolutional encoder–decoder architecture for image segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 12, pp. 2481–2495, Jan. 2017.
[62] Q. Zhu, C. Liao, H. Hu, X. Mei, and H. Li, "MAP-Net: Multiple attending path neural network for building footprint extraction from remote sensed imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 6169–6181, Jul. 2021.
[63] S. Ran, X. Gao, Y. Yang, S. Li, G. Zhang, and P. Wang, "Building multi-feature fusion refined network for building extraction from high-resolution remote sensing images," *Remote Sens.*, vol. 13, no. 14, p. 2794, Jul. 2021.
[64] P. Liu *et al.*, "Building footprint extraction from high-resolution images via spatial residual inception convolutional neural network," *Remote Sens.*, vol. 11, no. 7, p. 830, 2019.
[65] K. Chen, Z. Zou, and Z. Shi, "Building extraction from remote sensing images with sparse token transformers," *Remote Sens.*, vol. 13, no. 21, p. 4441, Nov. 2021.

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